

Engineering Mathematics

Previous Year Questions • 1993–2026

TRANSFORM THEORY

(EC, EE, IN, CS, ME, CE, CH, PI)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Transform Theory (EC/EE/IN/CS/ME/CE/CH/PI), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:





Easy



Medium



Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Branch Snapshot

A Maths unit can span up to eight branches (EC, EE, IN, CS, ME, CE, CH, PI), each weighted very differently in the real exam. Right after every Unit divider, a **Branch Snapshot** page lists every branch that unit covers, in the same order the questions themselves follow, each in its own compact box:

Questions	the real total for that branch.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy branch showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Branch Snapshot*

Branch: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real EC questions on this unit; on average EC has drawn about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** for your branch points to a consistently high-yield topic worth prioritizing; a low one is largely someone else's problem. A wide **Range** usually means the topic's importance has shifted over the years, so it is worth checking *which* years drove that before deciding how much weight to give it today.

Important Links & Resources

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

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UNIT

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Transform Theory

Topics

1. Laplace transform
   a) Basic Laplace Transforms
   b) Hyperbolic and Trigonometric Transforms
   c) Linearity Property
   d) Trigonometric Manipulations
   e) First Shifting Property
   f) Time division and multiplication
   g) Laplace Transform of Piecewise Functions
   h) Properties of Laplace Transform
   i) Inverse Laplace Transform
   j) Transform of Derivatives
   k) Solving Differntial Equation

Note to Reader

Transform theory are present directly or indirectly in branches with paper codes EC, EE, IN in subject Signal and System; MA, PH, ME, CE, CH, AE, BT, AG, MT, MN, PI, TF, BM, PE, CY in form of Laplace Transform according to GATE 2027 syllabus. Before starting please check with the latest syllabus.

Tranform theory for EC, EE, IN are solved extensively in Signal and System

Branches: ME,CE,CH,PI

Branch Snapshots*

ME — Mechanical Engineering

Questions: 16 **MCQ:** 16 **MSQ:** 0 **NAT:** 0
1M: 14 **2M:** 2 **Desc:** 0 **Avg Weightage** ~2M **Range:** 0M(2014)–2M(2026)

CE — Civil Engineering

Questions: 18 **MCQ:** 16 **MSQ:** 1 **NAT:** 1
1M: 14 **2M:** 4 **Desc:** 0 **Avg Weightage** ~1M **Range:** 1M(2025)–3M(2011)

CH — Chemical Engineering

Questions: 8 **MCQ:** 8 **MSQ:** 0 **NAT:** 0
1M: 5 **2M:** 3 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2020)–2M(2017)

PI — Production & Industrial Engineering

Questions: 3 **MCQ:** 3 **MSQ:** 0 **NAT:** 0
1M: 3 **2M:** 0 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2012)–1M(2012)

**See preface for how Avg Weightage and Range are calculated.*

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Mechanical Engineering

Q8.26.1 MCQ 2026 2M

Let $f(t)$ be a function of t defined for all positive values of t. The Laplace transform of $f(t)$ denoted by $L\{f(t)\} = \int_0^\infty e^{-st} f(t)\, dt$, provided that the integral exists where s is a parameter which may be a real or complex number. The Laplace transform of $f(t) = \sin 2t \sin 4t$ is

A. $\frac{16s}{(s^2+4)(s^2+36)}$ **B.** $\frac{32s}{(s^2+4)(s^2+36)}$

C. $\frac{16}{(s^2+4)(s^2+36)}$ **D.** $\frac{32}{(s^2+4)(s^2+36)}$

Q8.23.1 MCQ 2023 2M

Which one of the options given is the inverse Laplace transform of $\frac{1}{s^3 - s}$?

$u(t)$ denotes the unit-step function.

A. $(-1 + \frac{1}{2}e^{-t} + \frac{1}{2}e^{t})u(t)$

B. $(\frac{1}{2}e^{-t} - e^{t})u(t)$

C. $(-1 + \frac{1}{2}e^{-(t-1)} + \frac{1}{2}e^{(t-1)})u(t-1)$

D. $(-1 - \frac{1}{2}e^{-(t-1)} - \frac{1}{2}e^{(t-1)})u(t-1)$

Q8.22.1 MCQ 2022 1M

$F(t)$ is a periodic square wave function as shown. It takes only two values, 4 and 0, and stays at each of these values for 1 second before changing. What is the constant term in the Fourier series expansion of $F(t)$?

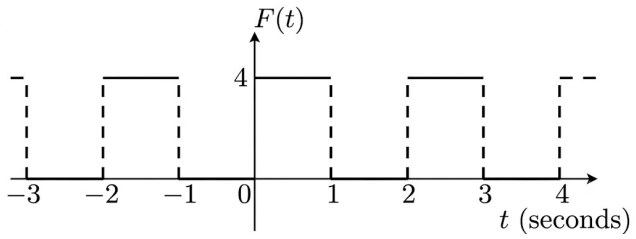


Fig. Q8.22.1

A. 1 **B.** 2 **C.** 3 **D.** 4

Q8.22.2 MCQ 2022 1M

The Fourier series expansion of x^3 in the interval $-1 \le x < 1$ with periodic continuation has

A. only sine terms

B. only cosine terms

C. both sine and cosine terms

D. only sine terms and a non-zero constant

Q8.15.1 MCQ 2015 1M

Laplace transform of $\cos(\omega t)$ is

A. $\frac{s}{s^2+\omega^2}$ **B.** $\frac{\omega}{s^2+\omega^2}$ **C.** $\frac{s}{s^2-\omega^2}$ **D.** $\frac{\omega}{s^2-\omega^2}$

Q8.15.2 MCQ 2015 1M

The Laplace transform of e^{i5t} where $i = \sqrt{-1}$, is

A. $\frac{s-5i}{s^2-25}$ **B.** $\frac{s+5i}{s^2+25}$ **C.** $\frac{s+5i}{s^2-25}$ **D.** $\frac{s-5i}{s^2+25}$

Q8.15.3 MCQ 2015 1M

Laplace transform of the function $f(t)$ is given by $f(s) = L\{f(t)\} = \int_0^\infty f(t)e^{-st}\, dt$. Laplace transform of the function shown below is given by

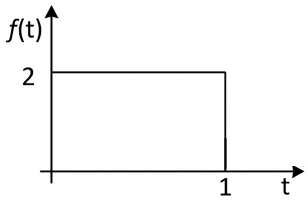


Fig. Q8.15.3

A. $\frac{1-e^{-2s}}{s}$ **B.** $\frac{1-e^{-s}}{s}$ **C.** $\frac{2-2e^{-s}}{s}$ **D.** $\frac{1-2e^{-s}}{s}$

Q8.15.4 MCQ 2015 1M

The Laplace Transform of $f(t) = e^{2t}\sin(5t)\, u(t)$ is

A. $\frac{5}{s^2-4s+29}$ **B.** $\frac{5}{s^2+5}$ **C.** $\frac{s-2}{s^2-4s+29}$ **D.** $\frac{5}{s+5}$

Q8.14.1 MCQ 2014 1M

Laplace transform of $\cos(\omega t)$ is $\frac{s}{s^2+\omega^2}$. The Laplace transform of $e^{-2t}\cos(4t)$ is

A. $\frac{s-2}{(s-2)^2+16}$ **B.** $\frac{s+2}{(s-2)^2+16}$

C. $\frac{s-2}{(s+2)^2+16}$ **D.** $\frac{s+2}{(s+2)^2+16}$

Q8.13.1 MCQ 2013 1M

The function $f(t)$ satisfies the differential equation $\frac{d^2f}{dt^2} + f = 0$ and the auxiliary conditions, $f(0) = 0, \frac{df}{dt}(0) = 4$. The Laplace transform of $f(t)$ is given by

A. $\frac{2}{s+1}$ **B.** $\frac{4}{s+1}$ **C.** $\frac{4}{s^2+1}$ **D.** $\frac{2}{s^4+1}$

Q8.12.1 MCQ 2012 1M

The inverse Laplace transform of the function $F(s) = \frac{1}{s(s+1)}$ is given by

A. $f(t) = \sin t$ **B.** $f(t) = e^{-t}\sin t$

C. $f(t) = e^{-t}$ **D.** $f(t) = 1 - e^{-t}$

Q8.10.1 MCQ 2010 1M

The Laplace transform of $f(t)$ is $\frac{1}{s^2(s+1)}$. The function is

A. $t - 1 + e^{-t}$ **B.** $t + 1 + e^{-t}$

C. $-1 + e^{-t}$ **D.** $2t + e^{t}$

Q8.07.1 MCQ 2007 1M

If $F(s)$ is the Laplace transform of the function $f(t)$ then Laplace transform of $\int_0^t f(x)\,dx$ is

A. $\frac{1}{s}F(s)$ **B.** $\frac{1}{s}F(s) - f(0)$

C. $sF(s) - f(0)$ **D.** $\int F(s)\,ds$

Q8.99.1 MCQ 1999 1M

Laplace transform of $(a+bt)^2$ where 'a' and 'b' are constants is given by:

A. $(a+bs)^2$

B. $1/(a+bs)^2$

C. $(a^2/s) + (2ab/s^2) + (2b^2/s^3)$

D. $(a^2/s) + (2ab/s^2) + (b^2/s^3)$

Q8.94.1 MCQ 1994 1M

If $f(t)$ is a finite and continuous Function for $t \geq 0$, the laplace transformation is given by $F = \int_0^\infty e^{-st} f(t)\,dt$. Then, for $f(t) = \cosh mt$, the laplace transformation is

A. $\frac{s}{s^2-m^2}$ **B.** $\frac{m}{s^2-m^2}$ **C.** $\frac{s}{s^2+m^2}$ **D.** $\frac{m}{s^2+m^2}$

Q8.93.1 MCQ 1993 1M

The laplace transform of the periodic function $f(t)$ described by the curve below i.e.

$$f(t) = \begin{cases} \sin t, & \text{if } (2n-1)\pi < t < 2n\pi \ (n = 1, 2, 3, \ldots) \\ 0 & \text{otherwise} \end{cases}$$

is

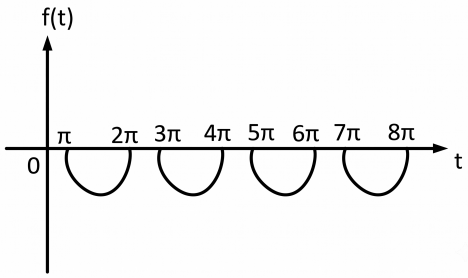


Fig. Q8.93.1

A. $\frac{e^{-\pi s}}{(1-s^2)(e^{-\pi s}-1)}$ **B.** $\frac{e^{-\pi s}}{(1+s^2)(e^{-\pi s}-1)}$

C. $\frac{e^{-\pi s}}{(1+s^2)(e^{-\pi s}+1)}$ **D.** $\frac{e^{\pi s}}{(1+s^2)(e^{-\pi s}-1)}$

Civil Engineering

Q8.25.1 MCQ 2025 1M

Which one of the following options is the correct Fourier series of the periodic function $f(x)$ described below:

$$f(x) = \begin{cases} 0 & \text{if } -2 < x < -1 \\ 2k & \text{if } -1 < x < 1; \text{ period} = 4 \\ 0 & \text{if } -1 < x < 2 \end{cases}$$

A. $f(x) = \frac{k}{2} + \frac{2k}{\pi}\left(\cos\frac{\pi}{2}x - \frac{1}{3}\cos\frac{3\pi}{2}x + \frac{1}{5}\cos\frac{5\pi}{2}x - + \ldots\right)$

B. $f(x) = \frac{k}{2} + \frac{2k}{\pi}\left(\sin\frac{\pi}{2}x - \frac{1}{3}\sin\frac{3\pi}{2}x + \frac{1}{5}\sin\frac{5\pi}{2}x - + \ldots\right)$

C. $f(x) = k + \frac{4k}{\pi}\left(\cos\frac{\pi}{2}x - \frac{1}{3}\cos\frac{3\pi}{2}x + \frac{1}{5}\cos\frac{5\pi}{2}x - + \ldots\right)$

D. $f(x) = k + \frac{4k}{\pi}\left(\sin\frac{\pi}{2}x - \frac{1}{3}\sin\frac{3\pi}{2}x + \frac{1}{5}\sin\frac{5\pi}{2}x - + \ldots\right)$

Q8.23.1 MSQ 2023 1M

The following function is defined over the interval $[-L, L]$:

$$f(x) = px^4 + qx^5$$

If it is expressed as a Fourier series,

$$f(x) = a_0 + \sum_{n=1}^{\infty}\left\{a_n \sin\left(\frac{n\pi x}{L}\right) + b_n \cos\left(\frac{n\pi x}{L}\right)\right\},$$

which options amongst the following are true?

A. $a_n,\ n = 1, 2, \ldots, \infty$ depend on p

B. $a_n,\ n = 1, 2, \ldots, \infty$ depend on q

C. $b_n,\ n = 1, 2, \ldots, \infty$ depend on p

D. $b_n,\ n = 1, 2, \ldots, \infty$ depend on q

Q8.22.1 NAT 2022 1M

The Fourier cosine series of a function is given by:

$$f(x) = \sum_{n=0}^{\infty} f_n \cos nx$$

For $f(x) = \cos^4 x$, the numerical value of $(f_4 + f_5)$ is _____ (round off to three decimal places).

Q8.11.1 MCQ 2011 1M

Given two continuous time signals $x(t) = e^{-t}$ and $y(t) = e^{-2t}$ which exists for $t > 0$ then the convolution $z(t) = x(t) * y(t)$ is

A. $e^{-t} - e^{-2t}$ **B.** e^{-2t} **C.** e^{-t} **D.** $e^{-t} + e^{-3t}$

Q8.11.2 MCQ 2011 2M

If $F(s) = L\{f(t)\} = \dfrac{2(s+1)}{s^2 + 4s + 7}$ then the initial and final values of $f(t)$ are respectively

A. 0, 2 **B.** 2, 0

C. 0, $\frac{2}{7}$ **D.** $\frac{2}{7}$, 0

Q8.09.1 MCQ 2009 1M

Laplace transform of $f(x) = \cosh(ax)$ is

A. $\dfrac{a}{s^2 - a^2}$ **B.** $\dfrac{s}{s^2 - a^2}$ **C.** $\dfrac{a}{s^2 + a^2}$ **D.** $\dfrac{s}{s^2 + a^2}$

Q8.05.1 MCQ 2005 2M

Laplace transform of $f(t) = \cos(pt + q)$ is

A. $\dfrac{s\cos q - p\sin q}{s^2 + p^2}$ **B.** $\dfrac{s\cos q + p\sin q}{s^2 + p^2}$

C. $\dfrac{s\sin q - p\cos q}{s^2 + p^2}$ **D.** $\dfrac{s\sin q + p\cos q}{s^2 + p^2}$

Q8.05.2 MCQ 2005 1M

The Laplace transform of a function $f(t)$ is

$$F(s) = \frac{5s^2 + 23s + 6}{s(s^2 + 2s + 2)}$$

As $t \to \infty$, $f(t)$ approaches

A. 3 **B.** 5 **C.** 17/2 **D.** ∞

Q8.04.1 MCQ 2004 1M

A delayed unit step function is defined as

$$u(t-a) = \begin{cases} 0, & t < a \\ 1, & t \geq a \end{cases}$$

Its Laplace transform is

A. $a\,e^{-as}$ **B.** e^{-as}/s **C.** e^{as}/s **D.** e^{-as}/a

Q8.03.1 MCQ 2003 1M

If L denotes the laplace transform of a function, $L\{\sin at\}$ will be equal to

A. $\dfrac{a}{s^2 - a^2}$ **B.** $\dfrac{a}{s^2 + a^2}$ **C.** $\dfrac{s}{s^2 + a^2}$ **D.** $\dfrac{s}{s^2 - a^2}$

Q8.02.1 MCQ 2002 2M

The Laplace transform of the following function is

$$f(t) = \begin{cases} \sin t & \text{for } 0 \leq t \leq \pi \\ 0 & \text{for } t > \pi \end{cases}$$

A. $1/(1 + s^2)$ for all $s > 0$

B. $1/(1 + s^2)$ for all $s < \pi$

C. $(1 + e^{-\pi s})/(1 + s^2)$ for all $s > 0$

D. $e^{-\pi s}/(1 + s^2)$ for all $s > 0$

Q8.01.1 MCQ 2001 1M

The inverse Laplace transform of $1/(s^2 + 2s)$ is

A. $(1 - e^{-2t})$ **B.** $(1 + e^{2t})/2$

C. $(1 - e^{2t})/2$ **D.** $(1 - e^{-2t})/2$

Q8.00.1 MCQ 2000 2M

Let $F(s) = L[f(t)]$ denote the Laplace transform of the function $f(t)$. Which of the following statements is correct?

A. $L[df/dt] = 1/s\,F(s)$; $L\left\{\int_0^t f(\tau)\,d\tau\right\} = sF(s) - f(0)$

B. $L[df/dt] = sF(s) - F(0)$; $L\left\{\int_0^t f(\tau)\,d\tau\right\} = -dF/ds$

C. $L[df/dt] = sF(s) - F(0)$; $L\left\{\int_0^t f(\tau)\,d\tau\right\} = F(s - a)$

D. $L[df/dt] = sF(s) - f(0)$; $L\left\{\int_0^t f(\tau)\,d\tau\right\} = \frac{1}{s}F(s)$

Q8.99.1 MCQ 1999 1M

The Laplace transform of the function

$$f(t) = k, \quad 0 < t < c$$
$$= 0, \quad c < t < \infty,$$

is

A. $(k/s)e^{-sc}$ **B.** $(k/s)e^{sc}$

C. $k\,e^{-sc}$ **D.** $(k/s)(1 - e^{-sc})$

Q8.98.1 MCQ 1998 1M

The Laplace Transform of a unit step function $u_a(t)$, defined as

$$u_a(t) = 0 \quad \text{for } t < a$$
$$= 1 \quad \text{for } t > a,$$

is

A. e^{-as}/s **B.** se^{-as} **C.** $s - u(0)$ **D.** $se^{-as} - 1$

Q8.98.2 MCQ 1998 1M

A discontinuous real function can be expressed as

A. Taylor's series and Fourier's series

B. Taylor's series and not by Fourier's series

C. Neither Taylor's series nor Fourier's series

D. not by Taylor's series, but by Fourier's series

Q8.96.1 MCQ 1996 1M

$(s + 1)^{-2}$ is laplace transform of

A. t^2 **B.** t^3 **C.** e^{-2t} **D.** te^{-t}

Q8.95.1 MCQ 1995 1M

The inverse Laplace transform of $\dfrac{(s + 9)}{(s^2 + 6s + 13)}$ is

A. $\cos 2t + 9 \sin 2t$

B. $e^{-3t} \cos 2t - 3e^{-3t} \sin 2t$

C. $e^{-3t} \sin 2t + 3e^{-3t} \cos 2t$

D. $e^{-3t} \cos 2t + 3e^{-3t} \sin 2t$

Chemical Engineering

Q8.20.1 MCQ 2020 1M

Consider the following unit step function.



Fig. Q8.20.1

The Laplace transform of this function is

A. $\dfrac{e^{-3s}}{s}$ **B.** $\dfrac{e^{-3s}}{s^2}$ **C.** $\dfrac{e^{-3s}}{3s}$ **D.** $\dfrac{e^{-6s}}{s}$

Q8.17.1 MCQ 2017 2M

The Laplace transform of function is $\dfrac{s + 1}{s(s + 2)}$. The initial and final values, respectively, of the function are

A. 0 and 1 **B.** 1 and $\frac{1}{2}$ **C.** $\frac{1}{2}$ and 1 **D.** $\frac{1}{2}$ and 0

Q8.16.1 MCQ 2016 1M

The Laplace transform of $e^{at} \sin(bt)$ is

A. $\dfrac{b}{(s - a)^2 + b^2}$ **B.** $\dfrac{(s - a)}{(s - a)^2 + b^2}$

C. $\dfrac{(s - a)}{(s - a)^2 - b^2}$ **D.** $\dfrac{b}{(s - a)^2 - b^2}$

Q8.14.1 MCQ 2014 1M

For the time domain function $f(t) = t^2$, which ONE of the following is the Laplace transform of $\int_0^t f(t)\,dt$?

A. $\frac{3}{s^4}$ **B.** $\frac{1}{4s^2}$ **C.** $\frac{2}{s^3}$ **D.** $\frac{2}{s^4}$

Q8.12.1 MCQ 2012 1M

The unilateral Laplace transform of $f(t)$ is $\frac{1}{s^2+s+1}$. The unilateral Laplace transform of $tf(t)$ is

A. $\frac{-s}{(s^2+s+1)^2}$ **B.** $\frac{-(2s+1)}{(s^2+s+1)^2}$

C. $\frac{s}{(s^2+s+1)^2}$ **D.** $\frac{2s+1}{(s^2+s+1)^2}$

Q8.08.1 MCQ 2008 2M

The Laplace transform of the function $f(t) = t\sin t$ is

A. $\frac{2s}{(s^2+1)^2}$ **B.** $\frac{1}{s^2(s^2+1)}$

C. $\frac{1}{s^2} + \frac{1}{(s^2+1)}$ **D.** $\frac{1}{(s-1)^2+1}$

Q8.07.1 MCQ 2007 2M

The Laplace transform of $f(t) = \frac{1}{\sqrt{t}}$ is

A. $\sqrt{\frac{\pi}{s}}$ **B.** $\frac{1}{\sqrt{s}}$ **C.** $\frac{1}{s^{3/2}}$ **D.** does not exist

Q8.98.1 MCQ 1998 1M

The Laplace transform of the function e^{-at} has the form:

A. $\frac{1}{s+1}$ **B.** $\frac{1}{s(s+a)}$ **C.** $\frac{a}{s}$ **D.** $\frac{1}{s+a}$

Production & Industrial Engineering

Q8.12.1 MCQ 2012 1M

The inverse Laplace transform of the function $F(s) = \frac{1}{s(s+1)}$ is given by

A. $f(t) = \sin t$ **B.** $f(t) = e^{-t}\sin t$

C. $f(t) = e^{-t}$ **D.** $f(t) = 1 - e^{-t}$

Q8.08.1 MCQ 2008 1M

Laplace transform of $\sin ht$ is

A. $\frac{1}{s^2-1}$ **B.** $\frac{1}{1-s^2}$ **C.** $\frac{s}{s^2-1}$ **D.** $\frac{s}{1-s^2}$

Q8.08.2 MCQ 2008 1M

Laplace transform of $8t^3$ is

A. $\frac{8}{s^4}$ **B.** $\frac{16}{s^4}$ **C.** $\frac{24}{s^4}$ **D.** $\frac{48}{s^4}$

Answer Key

MECHANICAL ENGINEERING					
Q8.26.1 A	Q8.23.1 A	Q8.22.1 B	Q8.22.2 A	Q8.15.1 A	Q8.15.2 B
Q8.15.3 C	Q8.15.4 A	Q8.14.1 D	Q8.13.1 C	Q8.12.1 D	Q8.10.1 A
Q8.07.1 A	Q8.99.1 C	Q8.94.1 A	Q8.93.1 B		
CIVIL ENGINEERING					
Q8.25.1 C	Q8.23.1 B, C	Q8.22.1 0.120 to 0.130	Q8.11.1 A	Q8.11.2 B	Q8.09.1 B
Q8.05.1 A	Q8.05.2 A	Q8.04.1 B	Q8.03.1 B	Q8.02.1 C	Q8.01.1 D
Q8.00.1 D	Q8.99.1 D	Q8.98.1 A	Q8.98.2 D	Q8.96.1 D	Q8.95.1 D
CHEMICAL ENGINEERING					
Q8.20.1 A	Q8.17.1 B	Q8.16.1 A	Q8.14.1 D	Q8.12.1 D	Q8.08.1 A
Q8.07.1 A	Q8.98.1 D				
PRODUCTION & INDUSTRIAL ENGINEERING					
Q8.12.1 D	Q8.08.1 A	Q8.08.2 D			

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